

Which Tail, and Probability of What?

Unit 3 · Topic 3.6 · THE PROBLEM

Suppose a city's earlier planning model says 55% of households support expanded evening bus service. A researcher suspects support has *fallen*. From the city's 20,000 households, a simple random sample of 200 finds 98 that support expansion, so $\hat{p} = 0.49$. Conditions for a one-sample z -test are met.

Priya: “For $H_0 : p = 0.55$ versus $H_a : p < 0.55$, use the left-tail p -value, about 0.044. If support is still 55%, about 4.4% of random samples of 200 would produce a sample proportion of 0.49 or lower.”

Noah: “Any difference from 0.55 counts, so double the tail to about 0.088. That means there is an 8.8% chance that the actual support proportion differs from 0.55.”

Sample counts

Group	Support	Other	Total
Households	98	102	200

FIRST TAKE — one sentence before you work the parts

Does fallen mean only a drop in support, or a change in either direction? Give one reason from the study.

- DOK 1** (a) State the null and alternative hypotheses and identify the tail direction required by the research question.
- DOK 2** (b) Compute the one-sample proportion z -statistic and its one-sided p -value. Also compute the two-sided p -value Noah would obtain.
- DOK 3** (c) Choose the warranted p -value and interpretation. Use H_a and $\hat{p} = 0.49$ to define the tail event and null assumption, then explain what Noah's doubled area answers and what his statement gets wrong.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.